AUTOMATED TRANSCRIPT SYSTEM

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Submitted for the partial fulfillment of BS Software Engineering degree to the Faculty of Engineering & Computing

NATIONAL UNIVERSITY OF MODERN LANGUAGES
ISLAMABAD AUGUST, 2024

ABSTRACT

This research addresses the critical challenges of data consistency, synchronization, and administrative throughput in large-scale academic institutions. The study focuses on the design and implementation of an **Automated Transcript System (ATS)**, transitioning from legacy manual workflows to a high-availability digital architecture specifically optimized for the National University of Modern Languages (NUML).

The research effort involved an exhaustive analysis of system requirements and architectural trade-offs. Adopting a **Structured Iterative Waterfall Lifecycle**, the development process was grounded in rigorous Engineering principles, moving from formal domain modeling to a centralized data repository architecture. Significant effort was directed toward designing a robust back-end using **ASP.NET MVC Core** and **MS SQL Server**, ensuring ACID (Atomicity, Consistency, Isolation, Durability) properties were maintained across complex relational data dependencies. This phase required an in-depth application of **Discrete Structures** and **Database Theory** to handle the multifaceted logic of multi-departmental transcript generation and real-time record evaluation.

A core component of this work is the **Systematic Validation Phase**, where **Black-Box testing** was integrated with **Selenium-based automation**. This allowed for the empirical evaluation of system performance and functional integrity under various simulated load scenarios. The research demonstrates that the application of modular software architecture significantly reduces latency in document generation while enhancing the scalability of academic record-keeping. The project concludes with an analysis of the system's ability to maintain data integrity within a high-traffic environment, providing a foundation for future integration of automated predictive analytics for student performance monitoring.